A Sharp One-Third Barrier for the $M^*(n, n)$ Route in the Numerical-Radius M-Ideal Question

Automatic math-discovery packet

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Source question

The source paper is Manwook Han and Sun Kwang Kim, *Geometry of the space of compact operators endowed with the numerical radius norm*, arXiv:2502.10821. For a Banach space X with positive numerical index $n(X)$, the paper asks:

> Let X be a Banach space with $n(X) > 0$. Is X Asplund whenever $\mathcal{K}_v(X)$ is an M-ideal in $\mathcal{L}_v(X)$?

The relevant source mechanism is Proposition 3.7: if X is infinite-dimensional, $n(X) > 0$, and $\mathcal{K}_v(X)$ is an M-ideal in $\mathcal{L}_v(X)$, then X has property $M^*(n(X), n(X))$. The source then uses the Cabello–Nieto–Oja theorem that the $M(r, s)$-inequality with $r + s > 1$ implies Asplundness, giving the Asplund conclusion when $n(X) > 1/2$. It also records that, under the same $n(X) > 1/2$ hypothesis, a space containing ℓ_1 cannot satisfy the numerical-radius M-ideal hypothesis.

Definitions

For $r, s \in (0, 1]$, a Banach space X has property $M^*(r, s)$ if, whenever $x^*, y^* \in X^*$, $\|x^*\| \leq \|y^*\|$, and (z^*_β) is a bounded w^*-null net in X^*, one has

$$\limsup_\beta \|rx^* + sz^*_\beta\| \leq \limsup_\beta \|y^* + z^*_\beta\|.$$

Idea of the Proof

The source proof loses the factor $n(X)$ when it converts numerical radius control back to operator-norm control. The resulting property $M^*(n, n)$ has its own intrinsic threshold: $M^*(r, r)$ is automatic for every Banach space as soon as $r \leq 1/3$. This follows from the triangle inequality and weak-star lower semicontinuity.

The constant is sharp. In $\ell_\infty = (\ell_1)^*$, take the constant functionals 1 and -1, and the weak-star-null sequence $2e_n$. Then $\|1 + 2e_n\|_\infty = 3$, while $\| - 1 + 2e_n\|_\infty = 1$. Thus $M^*(r, r)$ fails for every $r > 1/3$. A standard weak-star lifting argument transfers this obstruction from almost-isometric copies of ℓ_1 inside an arbitrary Banach space.

Main Result

Theorem 1 (Universal one-third bound)**.** *Every Banach space X has property $M^*(r, r)$ for every $0 < r \leq 1/3$. The same proof gives property $M(r, r)$ for every $0 < r \leq 1/3$.*

Proof. We prove the M^* statement. Let $x^*, y^* \in X^*$ satisfy $\|x^*\| \le \|y^*\|$, and let (z^*_β) be a bounded w^*-null net in X^*. Put

$$a = \|y^*\|, \qquad L = \limsup_\beta \|z^*_\beta\|.$$

Since $y^* + z^*_\beta \to y^*$ in the weak-star topology and the norm is weak-star lower semicontinuous,

$$\limsup_\beta \|y^* + z^*_\beta\| \ge a.$$

The triangle inequality also gives

$$\limsup_\beta \|y^* + z^*_\beta\| \ge L - a.$$

Hence

$$\limsup_\beta \|y^* + z^*_\beta\| \ge \max\{a, L - a\}.$$

For all $a, L \ge 0$,

$$\max\{a, L-a\} \ge \frac{a+L}{3}.$$

Indeed, if $L \le 2a$, then $a \ge (a+L)/3$; if $L \ge 2a$, then $L - a \ge (a+L)/3$. On the other hand,

$$\limsup_\beta \|x^* + z^*_\beta\| \le \|x^*\| + L \le a + L.$$

Therefore, for $0 < r \le 1/3$,

$$r \limsup_\beta \|x^* + z^*_\beta\| \le \frac{a+L}{3} \le \limsup_\beta \|y^* + z^*_\beta\|,$$

which is exactly $M^*(r,r)$. The proof of $M(r,r)$ is identical, replacing w^*-null nets in X^* by weakly null nets in X. □

Theorem 2 (Sharpness at ℓ_1)**.** *The space ℓ_1 fails property $M^*(r,r)$ for every $r > 1/3$. More generally, if a Banach space X has property $M^*(r,r)$ for some $r > 1/3$, then X contains no isomorphic copy of ℓ_1.*

Proof. First take $X = \ell_1$, so $X^* = \ell_\infty$ with its $\sigma(\ell_\infty, \ell_1)$ topology. Let

$$u = (1,1,1,\ldots), \qquad v = (-1,-1,-1,\ldots), \qquad z_n = 2e_n.$$

Then $\|u\|_\infty = \|v\|_\infty = 1$, and $z_n \to 0$ in $\sigma(\ell_\infty, \ell_1)$. Moreover

$$\|u + z_n\|_\infty = 3, \qquad \|v + z_n\|_\infty = 1$$

for every n. Thus $M^*(r,r)$ would force $3r \le 1$.

We now explain the standard lifting argument for arbitrary X. Suppose that X contains an isomorphic copy of ℓ_1. By James' distortion theorem for ℓ_1, for every $D > 1$ there is a subspace $E \subset X$ and an isomorphism $U : \ell_1 \to E$ such that

$$\|a\|_1 \le \|Ua\| \le D\|a\|_1 \qquad (a \in \ell_1).$$

For $b \in \ell_\infty$, let $\phi_b \in E^*$ be given by $\phi_b(Ua) = \sum_k b_k a_k$. Then

$$\|\phi_b\|_{E^*} \le \|b\|_\infty, \qquad \|\phi_b\|_{E^*} \ge D^{-1}\|b\|_\infty.$$

Let $b^+ = (1,1,\ldots)$, $b^- = (-1,-1,\ldots)$, and $c_m = b^- + 2e_m$. The functionals ϕ_{b^+} and ϕ_{b^-} have the same norm. Choose norm-preserving Hahn–Banach extensions $u, v \in X^*$ of ϕ_{b^+} and ϕ_{b^-}, respectively.

We use the following elementary finite-dimensional form of weak-star lifting. For every finite-dimensional $F \subset X$ and every $\varepsilon > 0$, for all sufficiently large m there exists $h_{m,F,\varepsilon} \in X^*$ extending ϕ_{c_m} such that

$$\|h_{m,F,\varepsilon}\| \le 1 + \varepsilon, \qquad |h_{m,F,\varepsilon}(x) - v(x)| \le \varepsilon\|x\| \quad (x \in F).$$

Indeed, $\phi_{c_m} - \phi_{b^-} = 2\phi_{e_m}$ converges pointwise to 0 on E, hence uniformly on the unit ball of the finite-dimensional space $E \cap F$. The two prescriptions ϕ_{c_m} on E and $v|_F$ are therefore compatible up to ε on $E \cap F$; the usual finite-dimensional Hahn–Banach perturbation lemma gives an extension on $E + F$, and Hahn–Banach extends it to X.

Set $z_{m,F,\varepsilon} = h_{m,F,\varepsilon} - v$, with the directed index (F, ε) ordered by inclusion of F and decreasing ε, and m chosen large enough as above. Then $(z_{m,F,\varepsilon})$ is a bounded w^*-null net in X^*. Furthermore,

$$\limsup \|v + z_{m,F,\varepsilon}\| = \limsup \|h_{m,F,\varepsilon}\| \le 1,$$

while the restriction of $u + z_{m,F,\varepsilon}$ to E is $\phi_{b^+} - \phi_{b^-} + \phi_{c_m} = \phi_{b^+ + 2e_m}$, so

$$\limsup \|u + z_{m,F,\varepsilon}\| \ge D^{-1}\|b^+ + 2e_m\|_\infty = 3D^{-1}.$$

If X had property $M^*(r,r)$, we would get $3rD^{-1} \le 1$, i.e. $r \le D/3$. Since $D > 1$ is arbitrary, $r \le 1/3$. Hence no space with property $M^*(r,r)$, $r > 1/3$, can contain an isomorphic copy of ℓ_1. □

Corollary 1 (Improved ℓ_1-exclusion for the source question)**.** *Let X be a Banach space with $n(X) > 1/3$. If $\mathcal{K}_v(X)$ is an M-ideal in $\mathcal{L}_v(X)$, then X contains no isomorphic copy of ℓ_1.*

Proof. By Proposition 3.7 of Han–Kim, the numerical-radius M-ideal hypothesis implies property $M^*(n(X), n(X))$. Since $n(X) > 1/3$, Theorem 2 excludes every isomorphic copy of ℓ_1. □

Corollary 2 (Complex spaces)**.** *If X is complex and $\mathcal{K}_v(X)$ is an M-ideal in $\mathcal{L}_v(X)$, then X contains no isomorphic copy of ℓ_1.*

Proof. The standard Bonsall–Duncan lower bound gives $n(X) \ge 1/e > 1/3$ for every complex Banach space. Apply Corollary 1. □

Limitations

This packet does not prove that the source question has a positive answer. The new conclusion rules out ℓ_1-containment under the stronger threshold $n(X) > 1/3$, but Asplundness is stronger than not containing ℓ_1. In particular, James-tree-type mechanisms show why non-Asplund spaces without ℓ_1 remain a genuine obstacle.

The packet also shows a hard limitation of the source paper's immediate route: for $n(X) \le 1/3$, property $M^*(n(X), n(X))$ is automatic for every Banach space, so it cannot by itself imply Asplundness.

Novelty and verification notes

The cheap run indexes had no exact prior packet for arXiv:2502.10821 or this numerical-radius M-ideal question. A bounded web search for the exact question and for phrases involving $M^*(r,s)$, 1/3, and ℓ_1 found the source paper but no later answer or matching constant-barrier result.

Human review should focus on the finite-dimensional lifting paragraph in Theorem 2; the remaining estimates are scalar norm calculations. No numerical computation was used.

References

[1] F. F. Bonsall and J. Duncan, *Numerical ranges of operators on normed spaces and of elements of normed algebras*, London Mathematical Society Lecture Note Series, vol. 2, Cambridge University Press, 1971.

[2] J. C. Cabello, E. Nieto, and E. Oja, *On ideals of compact operators satisfying the $M(r, s)$-inequality*, J. Math. Anal. Appl. 220 (1998), 334–348.

[3] M. Han and S. K. Kim, *Geometry of the space of compact operators endowed with the numerical radius norm*, arXiv:2502.10821.

[4] R. C. James, *Uniformly non-square Banach spaces*, Ann. of Math. (2) 80 (1964), 542–550.